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Ferroelastic altermagnetism

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Combination of altermagnetism and ferroic orders, such as ferroelectric switchable altermagnetism [*Phys. Rev. Lett.* **134**, 106801 (2025) and *Phys. Rev. Lett.* **134**, 106802 (2025)], offers a powerful route to achieve nonvolatile switching of altermagnetic spin splitting. In this work, by synergizing altermagnetism and ferroelasticity, we propose the concept of *ferroelastic altermagnets* in which the ferroelastic crystal reorientation can drive multistate nonvolatile switching of the altermagnetic spin splitting via *altermagnetoelastic effect*. Using monolayers RuF₄ and CuF₂ as material candidates, we demonstrate 2-state and 3-state altermagnetic spin splitting switching as driven by ferroelastic strain states. Transport calculation shows that multistate spin conductivities can be ferroelastically encoded in ferroelastic altermagnets, thus suggesting the potential of ferroelastic altermagnets as nonvolatile nanomechanical spin switches. The proposed concept of ferroelastic altermagnetism enriches the emerging landscape of multiferroic altermagnetism and shall pave a way towards straintronic-spintronic device applications.

Together with ferromagnets (FM) and conventional antiferromagnets (AFM), altermagnets represent an emerging new family of collinear magnetic materials^{1–5}. Altermagnets exhibit zero net magnetization like conventional AFM, but host nonrelativistic band spin splitting like FM. Such FM-AFM dichotomy endows altermagnets with the advantages of AFM such as robustness against external fields and compatible with ultrahigh-speed device operation, while still exhibiting broken spin degeneracy of FM, which are critical for spin manipulation and information processing^{6–24}. Altermagnets exhibit a plethora of intriguing physical phenomena such as spin current generation^{1,2}, spin Hall effect⁶, spin Nernst effect⁷, anomalous Hall effect⁸, tunneling magnetoresistance effects⁶, and proximity effect²⁵, which can be harnessed for various device applications. A large variety of altermagnets have been theoretically proposed^{13,4,17,26}, and some of them have been experimentally confirmed^{27–30}.

The physics of altermagnets can be further enriched by *multiferroicity*. Multiferroics are singular materials that exhibit two or more ferroic orders among (anti)ferromagnetism, (anti)ferroelectricity and (anti)ferroelasticity^{31–39}. For instance, the coupling between ferroelectricity and magnetism yields *magnetoelectric effect* where ferroelectric polarization and magnetization are coupled and can be mutually switched. Recent studies have unveiled the altermagnetic counterpart of magnetoelectric effect, namely the *altermagnetolectric effect* [Fig. 1a] in which the altermagnetic spin splitting can be tuned by various forms of ferroelectric switching^{40–48}, thus unveiling a route towards electrical-based nonvolatile switching of altermagnetism. The multiferroic integration of ferroelectricity and altermagnetism immediately raises the following question: Can altermagnetism

be integrated with *ferroelasticity*^{49–53}—*magnetoelastic effect*—to achieve nonvolatile *nanomechanical* deformation-induced switching of the altermagnetic spin splitting?

Here we propose the concept of *ferroelastic altermagnetism* in which altermagnetism and ferroelasticity are synergized. As ferroelastic switching is equivalent to a rotation operation on the lattice, such lattice rotation changes the spin-momentum coupling in a ferroelastic altermagnet, thus enabling an *altermagnetoelastic effect* in which the altermagnetic spin splitting can be mechanically switched [Fig. 1b, c]. Using monolayers RuF₄ and CuF₂ as proof-of-concept, 2-state and 3-state nonvolatile switching of the altermagnetic spin splitting are demonstrated, respectively, via first-principles calculations. We further show that the multistate ferroelastic switching lead to nonvolatile switchable spin transport. These findings reveal a previously unexplored mechanism to achieve nonvolatile strain-based altermagnetic switching that enables *multistate* nonvolatile information encoding. Ferroelastic altermagnetism represents a *mechanical* multiferroic counterpart of the recently proposed ferroelectric altermagnetism^{40–45}. In ferroelectric altermagnets, the reversal of electric polarization switches the altermagnetic spin splitting via internal electric-field control of the electronic structure, typically resulting in binary switching. In contrast, ferroelastic altermagnets utilize lattice reorientation to modify the altermagnetic symmetry, allowing multistate and mechanically driven nonvolatile control. Such strain-driven manipulation is particularly appealing for integrating altermagnetism into straintronic-spintronic platforms, where mechanical deformation can directly encode spin-

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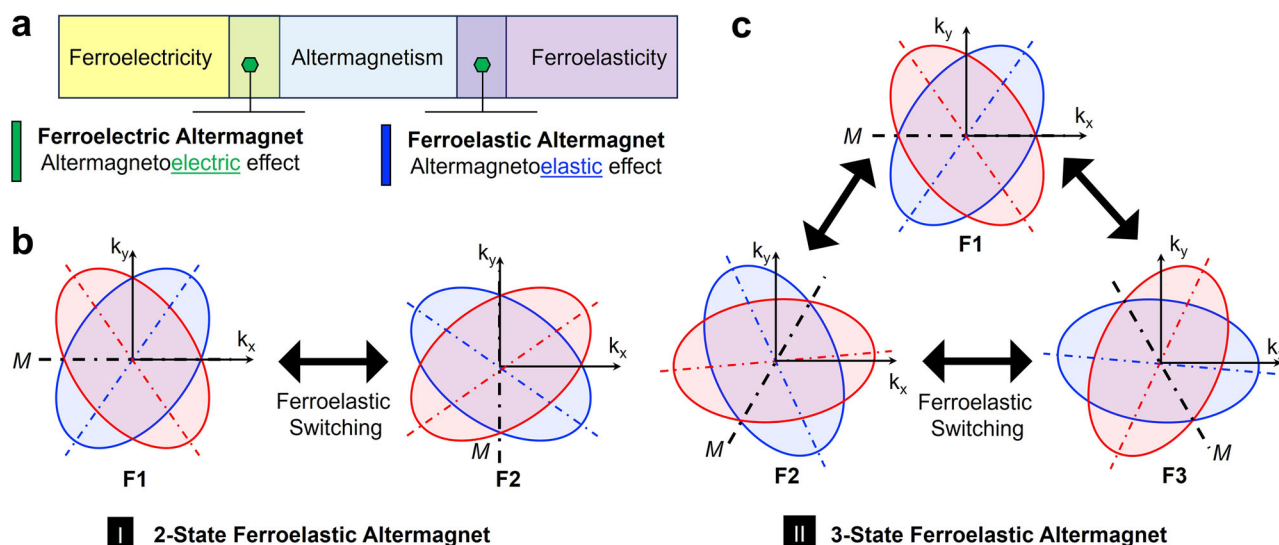


Fig. 1 | Concept of ferroelastic altermagnetism. **a** Ferroelastic altermagnets arise in a material where altermagnetism and ferroelasticity are simultaneously present. Schematic diagram of **b** 2-state and **c** 3-state nonvolatile mechanical switching of the

altermagnetic spin splitting in two-dimensional ferroelastic altermagnets. *M* is the mirror symmetry axis of the systems.

dependent functionalities. Notice that the piezomagnetic effect describes the linear coupling between magnetization (*M*) and stress (σ), or conversely, between strain (ϵ) and a magnetic field (*H*), expressed as $M_i = Q_{ijk}\sigma_{jk}$ or $\epsilon_{jk} = Q_{ijk}H_i$ ². In contrast, the magnetoelastic effect investigated in this work operates through a fundamentally different, nonlinear mechanism. Here, an applied stress does not primarily induce magnetization via a linear response but instead drives a ferroelastic phase transition, which in turn alters the altermagnetic order and the spin splitting, ultimately leading to the generation of spin currents with varying magnitudes and directions. Consequently, the relationship between stress and the altermagnetic order is nonlinear, governed by: $F = B_{ijkl} (M_i M_j) \epsilon_{kl}$, where *F* is free energy.

Results

Ferroelastic-driven altermagnetic spin switching and material candidates

Two-dimensional (2D) materials can be classified according to layer groups⁵⁴, and those belonging to layer groups L_8 – L_{48} may host ferroelasticity, as these groups often correspond to rectangular 2D Bravais lattices. Altermagnetic monolayers within these layer groups may thus exhibit coexisting ferroelasticity and altermagnetism, as both phenomena are symmetry-allowed in this regime¹⁸. To illustrate the concept of ferroelastic altermagnetism, we use monolayers RuF₄ and CuF₂ (layer group L_{17}) as proof-of-concept for achieving 2-state and 3-state ferroelastic-driven altermagnetic spin switching. Monolayers RuF₄ and CuF₂ can potentially be mechanically exfoliated from bulks RuF₄ and CuF₂ due to their layered structures^{55,56}. To confirm this point, we calculate their exfoliation energies. The exfoliation energies of monolayers RuF₄ and CuF₂ are 18 and 17 meV/Å², respectively, which are comparable to those of graphene (21 meV/Å²) and MoS₂ (18 meV/Å²)⁵⁷. As shown in Fig. 2a, the crystal lattice of monolayer RuF₄ has a rectangular lattice with a space group of $P2_1/c$. The Ru atoms are encapsulated in a distorted RuF₆ octahedron, and each Ru atom is connected to six F atoms. The optimized lattice constants of monolayer RuF₄ are $a = 5.42$ Å and $b = 5.08$ Å. The crystal structure of monolayer CuF₂ shares the same space group $P2_1/c$ with monolayer RuF₄ [Fig. 2b]. Also, the Cu atoms are encapsulated in a distorted CuF₆ octahedron, and each Cu atom is connected to six F atoms. The difference is that the adjacent RuF₆ octahedrons are connected via the corner-sharing manner. The optimized lattice constants of monolayer CuF₂ are $a = 5.32$ Å and $b = 3.56$ Å. The stabilities of

monolayers RuF₄ and CuF₂ are confirmed by phonon and ab initio molecular dynamics simulations (Fig. S1⁵⁸).

Under the octahedral crystal field, the d orbitals split into t_{2g} triplets and e_g doublets. In monolayer RuF₄, Ru atoms lose four electrons and transform into Ru⁴⁺ ions with four d electrons. Two of them occupy one t_{2g} orbital, one of them occupies another t_{2g} orbital, and one of them occupies the remaining t_{2g} orbital, giving rise to a local magnetic moment of $2\mu_B$. The inequivalent distribution of the electrons on the three t_{2g} orbitals leads to the compressed Jahn–Teller distortions of the crystal structure (axial bonds: 1.85 Å, equatorial bonds: 2.01 Å), which further splits the t_{2g} and e_g orbitals (Fig. S2⁵⁸). In monolayer CuF₂, Cu atoms lose two electrons and transform into Cu²⁺ ions with nine 3d electrons. Six of them occupy the three t_{2g} orbitals, two of them occupy one e_g orbital, and one of them occupies the other e_g orbital, giving rise to a local magnetic moment of $1\mu_B$. The inequivalent distribution of the electrons on the two e_g orbitals leads to the elongated Jahn–Teller distortions of the crystal structure (axial bonds: 2.62 Å, equatorial bonds: 1.93 Å), making t_{2g} and e_g orbitals split in a different way with those in monolayer RuF₄ (Fig. S2⁵⁸). The calculated magnetic moments on Ru and Cu atoms are 1.67 and 0.64 μ_B , respectively, which is consistent with the above analysis.

The magnetic ground states of monolayers RuF₄ and CuF₂ are determined by comparing the energy among three magnetic configurations: FM, Néel antiferromagnetic (AFM–Néel), and stripe antiferromagnetic (AFM–stripe) (Fig. S3⁵⁸). It is found that the energies of FM and AFM–stripe are 30.28 (7.47) and 11.69 (4.27) meV/unit cell higher than the AFM–Néel in monolayer RuF₄ (CuF₂), respectively, thus indicating AFM–Néel is the energetically most favorable state for both RuF₄ and CuF₂. The spin charge densities of AFM–Néel RuF₄ and CuF₂ are shown in Fig. 2c, d. After considering magnetism, the two opposite-spin sublattices in RuF₄ and CuF₂ can only be transformed to each other by screw or glide symmetry, which is typical for altermagnets.

Monolayer RuF₄ has an indirect band gap of 0.82 eV, with CBM located at *M*/*M'* point and VBM located along Γ –*M*/*M'* line [Fig. 2e]. The spin-up and spin-down states are degenerate along *M*–*X*– Γ –*Y*–*M*, while an alternating spin splitting appears along *M'*– Γ –*M*, forming a spin-momentum locking physics. We also plotted the two highest valence bands and the amplitude of the spin splitting ΔE between them over the entire Brillouin zone. The spin-momentum locking physics can also be observed at generic *k* points. The amplitude of the spin splitting reaches up to 205 meV. For monolayer CuF₂, its indirect band gap is 0.69 eV, with

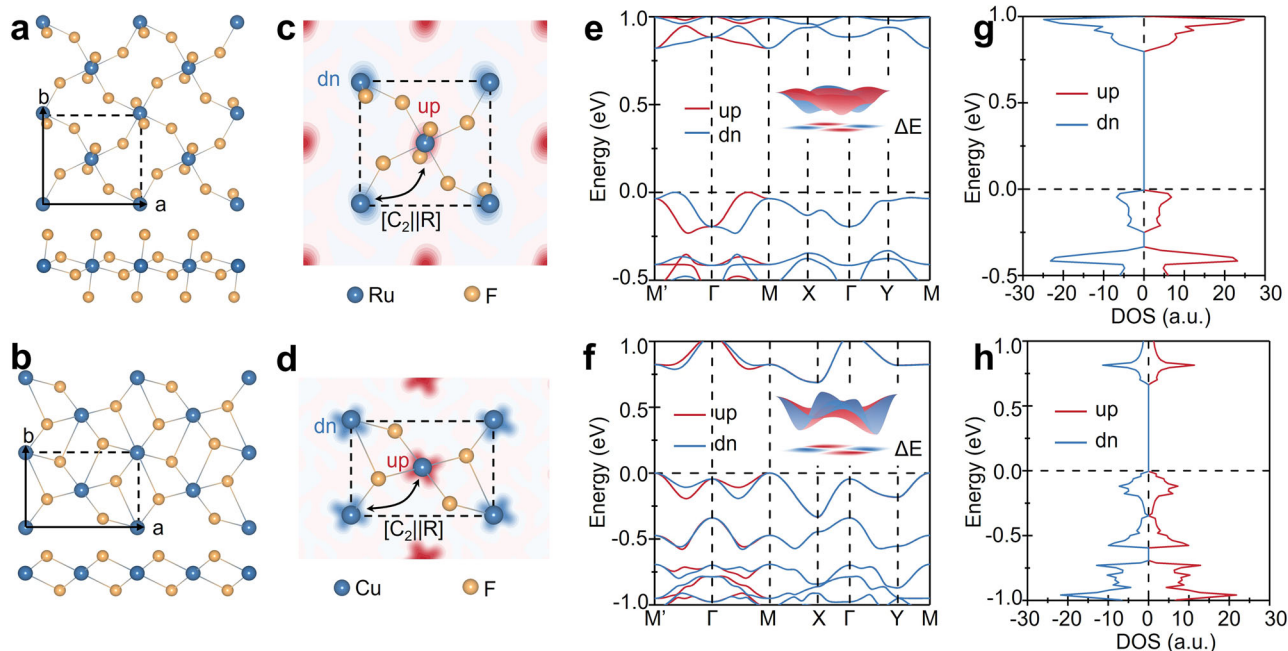


Fig. 2 | Crystal and electronic structures of 2D ferroelastic altermagnets. Crystal structures of **a** RuF₄ and **b** CuF₂ from top and side views. Spin charge densities of **c** RuF₄ and **d** CuF₂. Red and blue shading represent spin up and down, respectively.

Band structures of **e** RuF₄ and **f** CuF₂. Insets are two highest valence bands, and alternating spin splitting ΔE between them in the hole Brillouin zone. Density of states of **g** RuF₄ and **h** CuF₂. Fermi level is set to zero in (e–h).

CBM located at X point and VBM located at M/M' point [Fig. 2f]. Similarly, the spin-up and spin-down states are degenerate along M-X- Γ -Y-M, while an alternating spin splitting appears along M'- Γ -M. The amplitude of the spin splitting reaches up to 98 meV. The spin-up and spin-down density of states of RuF₄ and CuF₂ are equal [Fig. 2g, h].

Symmetry analysis

The underlying mechanism of the spin-momentum locking in monolayers RuF₄ and CuF₂ can be elucidated through symmetry analysis. When omitting the nonmagnetic F atoms, the systems have PT and $t_{1/2}T$ symmetry, thus guaranteeing that the band structures are spin-degenerate over the entire Brillouin zone. The F atoms in the non-centrosymmetric sites break the PT and $t_{1/2}T$ symmetry, yielding spin splitting in the band structures. The space group of monolayers RuF₄ and CuF₂, $P2_1/c$, contains four symmetry operations: {E, P, $t_{1/2}M_y$, $t_{1/2}C_{2y}$ }. The former two connects the same-spin sublattices, while the latter two connects the opposite-spin sublattices. Considering the crystal and spin rotation symmetries remain decoupled when considering AFM coupling in the absence of spin-orbital coupling, the symmetry connecting two opposite-spin sublattices can be represented as $[C_2||R]$, where the transformations on the left (right) of the double vertical bar act only on the spin (real) space. When the $[C_2||t_{1/2}M_y]$ and $[C_2||t_{1/2}C_{2y}]$ is operated on monolayer CuF₂, respectively, we obtain:

$$\begin{aligned} [C_2||t_{1/2}M_y]E(k_x, k_y, \uparrow) &= E(k_x, -k_y, \downarrow) \\ [C_2||t_{1/2}C_{2y}]E(k_x, k_y, \uparrow) &= E(-k_x, k_y, \downarrow) \end{aligned} \quad (1)$$

Consequently, we have

$$\begin{aligned} E(k_x, k_y, \uparrow) &= E(k_x, -k_y, \downarrow) \\ E(k_x, k_y, \uparrow) &= E(-k_x, k_y, \downarrow) \end{aligned} \quad (2)$$

This explains the emergence of alternating spin-split bands along M'- Γ -M and at generic k points. We also have

$$\begin{aligned} E(k_x, 0, \uparrow) &= E(k_x, 0, \downarrow) \\ E(0, k_y, \uparrow) &= E(0, k_y, \downarrow) \end{aligned} \quad (3)$$

After considering the Bloch theorem, the bands are spin-degenerate when $k_x = m\pi$ and $k_y = n\pi$, thus explaining the spin-degenerate bands along high-symmetry paths M-X- Γ -Y-M.

Multistate nonvolatile ferroelastic-altermagnetic spin splitting switching

Monolayer RuF₄ (CuF₂) has four independent elastic constants, which are $C_{11} = 39.46$ (56.96) N m⁻¹, $C_{12} = 20.25$ (9.76) N m⁻¹, $C_{22} = 31.47$ (19.83) N m⁻¹, and $C_{66} = 33.24$ (14.26) N m⁻¹. They meet the Born criteria: $C_{11}C_{22} - C_{12}^2 > 0$ and $C_{66} > 0$, indicating that RuF₄ (CuF₂) is mechanically stable. On the basis of the elastic constants, the Young's modulus of RuF₄ (CuF₂) can be obtained by the following expression⁵⁹:

$$Y(\theta) = \frac{C_{11}C_{22} - C_{12}^2}{C_{22}\cos^4\theta + A\cos^2\theta\sin^2\theta + C_{11}\sin^4\theta} \quad (4)$$

Here, $A = (C_{11}C_{22} - C_{12}^2)/C_{66} - 2C_{12}$. As shown in Fig. S4⁵⁸, the Young's modulus of monolayer RuF₄ (CuF₂) varies from 21 (18) N m⁻¹ to 61 (52) N m⁻¹, suggesting the significant mechanical anisotropy. These values are smaller than those of phosphorene (24–102 N m⁻¹), graphene (340 N m⁻¹) and MoS₂ (120–240 N m⁻¹)⁶⁰. The small Young's modulus ensures their mechanical flexibility, which is beneficial for the ferroelastic transition. RuF₄ has two energy-equivalent ferroelastic states F1 ($a > b$) and F2 ($a < b$), and one paraelastic state P ($a = b$) [Fig. 3a]. The ferroelastic switching from F1 to F2 is equivalent to a $\pi/2$ rotation of the lattice. In contrast, CuF₂ has three ferroelastic states. For convenience, we choose the deformed hexagon to be the structure unit. The three energy-equivalent ferroelastic phases can be described as F1 ($a_1 > a_2/a_3$), F2 ($a_2 > a_1/a_3$) and F3 ($a_3 > a_1/a_2$), where a_1 , a_2

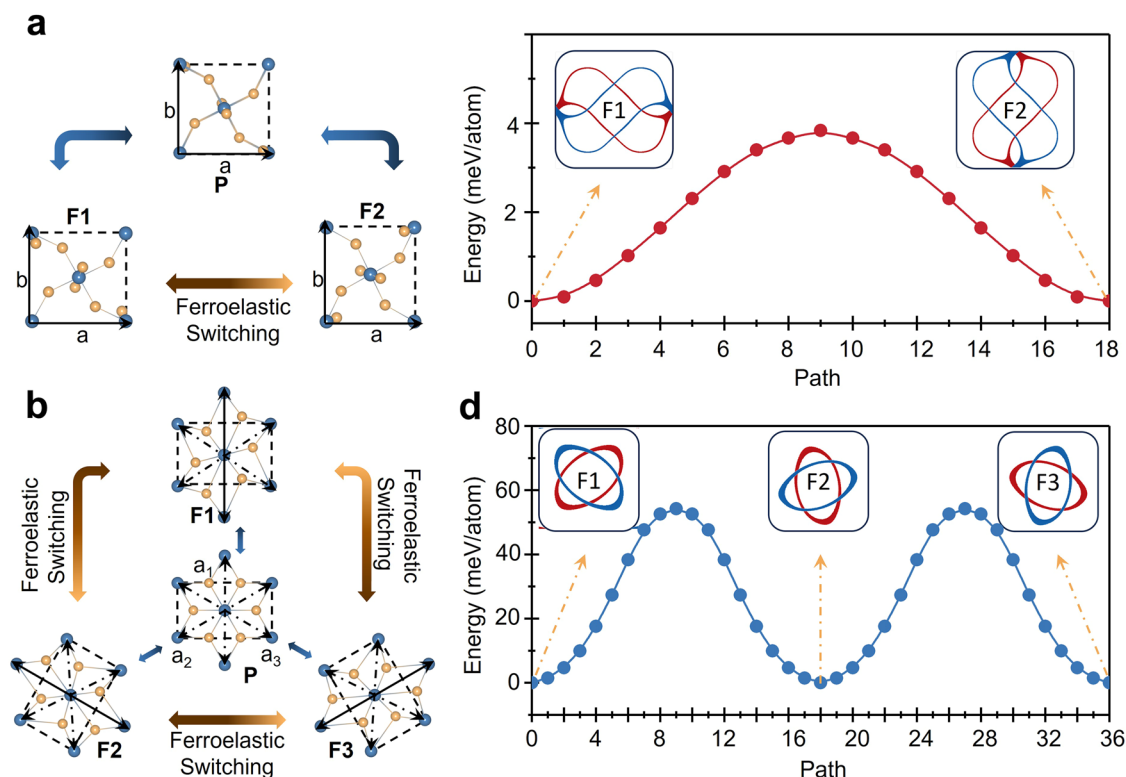


Fig. 3 | Ferroelastic switching in RuF₄ and CuF₂. Schematic diagrams of 2-state and 3-state ferroelastic switching in **a** RuF₄ and **b** CuF₂, respectively. Here a_1 , a_2 and a_3 are three diagonal lines of the deformed hexagon of CuF₂. Energy profiles of the

ferroelastic transition for **c** RuF₄ and **d** CuF₂ as a function of step number in NEB calculations. Insets show the spin-momentum locking in different ferroelastic phases at -0.1 eV below the Fermi level.

and a_3 are the diagonal lines of the deformed hexagon [Fig. 3b]. The paraelastic phase P is the T-phase CuF₂ with $a_1 = a_2 = a_3$. The ferroelastic switching from F1 to F2 (F3) is equivalent to a $2\pi/3$ ($4\pi/3$) lattice rotation. The ferroelastic reversible strain of RuF₄ and CuF₂, which are defined as $(a/b - 1) \times 100\%$ and $(a_i/a_j - 1) \times 100\%$, are calculated to be 6.8% and 11.4%, respectively, which are larger than those of SnS (4.9%) and SnSe (2.1%)⁴⁹, but smaller than those of borophane (42%)⁵⁰ and phosphorene (37.9%)⁴⁹. Given their moderate ferroelastic reversible strain and mechanical flexibility, the ferroelastic transition in RuF₄ and CuF₂ should be realistically achievable.

We simulate the ferroelastic switching process of RuF₄ and CuF₂ using the nudged elastic band (NEB) method. The overall switching barrier of RuF₄ and CuF₂ are 4 and 54 meV/atom [Fig. 3c, d], respectively, which is lower than those of phosphorene (200 meV/atom)⁴⁹ and borophane (100 meV/atom)⁵⁰. The low switching barrier suggests that monolayers RuF₄ and CuF₂ are potentially compatible with fast and low-energy ferroelastic switching operation. For RuF₄, the magnetic ground state maintains altermagnetism during the whole ferroelastic transition. For CuF₂, the magnetic ground state turns from altermagnetism to FM when F1 transforms to P, and then turn back to altermagnetism when P transforms to F2/F3. Upon ferroelastic switching, the reoriented lattice changes spin-momentum locking fashion, thus realizing multistate nonvolatile switching of the altermagnetic spin splitting [see insets in Fig. 3c, d].

In altermagnets, the spin-polarized current can be generated by an external in-plane electric field due to the anisotropic spin splitting^{1,2}. In ferroelastic altermagnets, the spin transport current can be mechanically switched via different ferroelastic states. We calculate the transport current in RuF₄ and CuF₂ when driven by an external field⁵⁸. The longitudinal and transverse spin conductivities exhibit an angular dependence on φ with a period of π , where φ is the direction of the in-plane electric field relative to x axis [Fig. 4a] and φ can be experimentally controlled by changing the relative angle between the source-drain electrode axis [see Fig. 4b for a device schematic] and the crystal structure of ferroelastic altermagnet sample. Upon a ferroelastic switching, the lattice re-orientation changes the relative

orientation between the altermagnetic spin-split band structure and the electric field, thus leading to spin current switching effect. In RuF₄, the carrier transport under F1 and F2 states exhibits equal but opposite transverse and longitudinal spin conductivities [Fig. 4c, d]. The 2-state ferroelastic switching between F1 and F2 states thus lead to spin reversal. For CuF₂, the spin conductivities exhibit even richer switching effect via the three ferroelastic states F1, F2 and F3. For instance, when the electric field is directed at $\varphi = 0$, the longitudinal spin conductivities under F1, F2 and F3 states can be switched between positive, zero and negative magnitudes, respectively [Fig. 4e, f]. The ferroelastic altermagnetic broadens the application scenario of altermagnets, enabling nonvolatile spin information encoding and hybrid straintronic-spintronic device application.

To examine the effects of spin-orbital coupling (SOC), we perform calculations considering SOC. The magnetic orientations of RuF₄ and CuF₂ can be determined by magnetic anisotropy energy (MAE). The MAE are 4.5 meV/Ru and -0.3 meV/Cu, indicating that the magnetic easy axes of RuF₄ and CuF₂ are along the x and z direction, respectively. Then the magnetic and spin space groups of RuF₄ (CuF₂) are $P2_1'/c'$ and $P^{-1}2_1'/1^{-1}c^{-m}1$ ^{61,62}, which allows weak ferromagnetism. Our calculated results show that the spins of Ru (Cu) ions tilt from x to y (z to y), resulting in a weak magnetic moment $\sim 0.11 \mu_B/\text{Ru}$ ($\sim 0.03 \mu_B/\text{Cu}$) along y . The band structures with SOC are shown in Fig. S5⁵⁸. As we can see, although the dispersion of the energy bands is somewhat influenced by SOC, they largely retain their original spin splitting. Consequently, the resulting spin current is affected to some extent, but the underlying mechanism remains applicable, as seen in the case of RuO₂¹. In practical applications, dynamic strain can be induced through relevant experimental methods including piezoelectric strain cells for low-frequency dynamic strain⁶³, dynamic mechanical analyzers for low-frequency oscillatory strain⁶⁴, and surface acoustic waves for MHz–GHz strain modulation⁶⁵. These techniques have been successfully used to realize dynamic and reversible strain control in a variety of material systems, demonstrating the experimental feasibility of dynamic strain modulation in our manuscript. Each distinct strain states corresponds to a unique spin conductivity in the

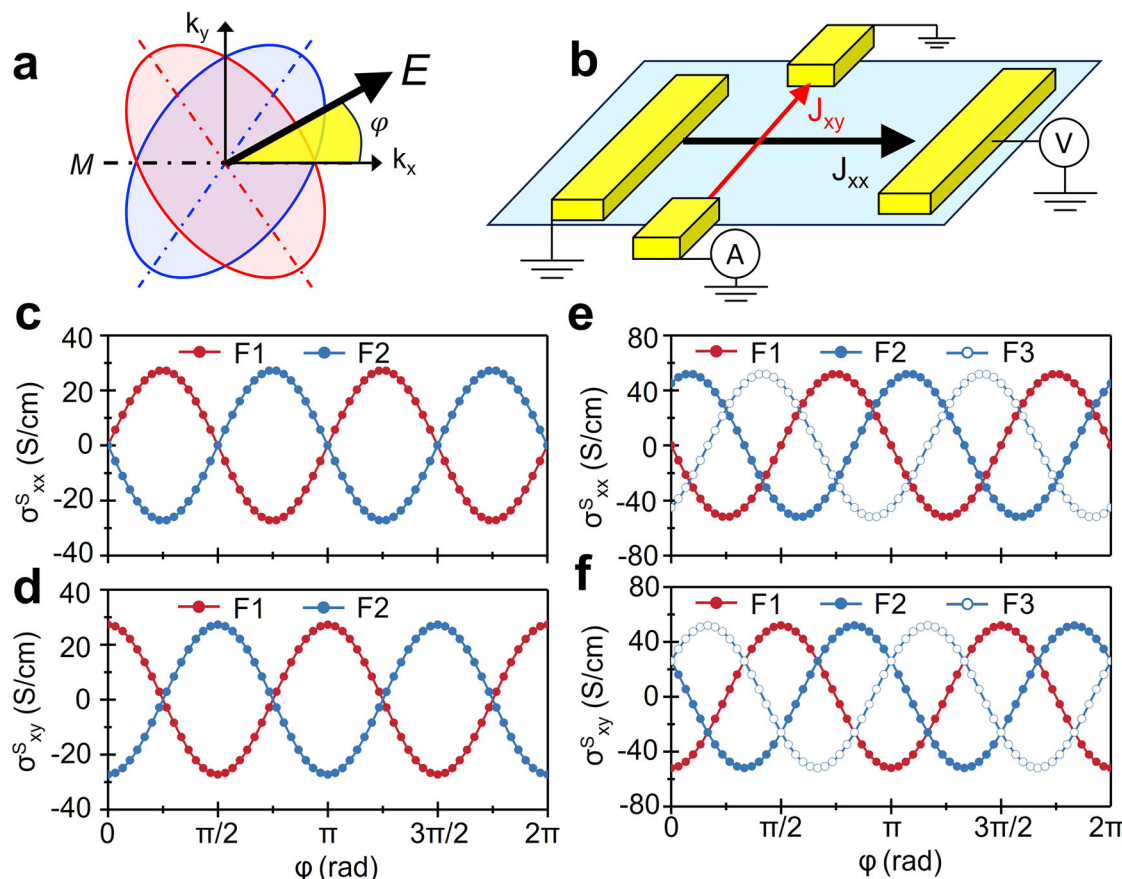


Fig. 4 | Transport properties in RuF₄ and CuF₂. **a** Schematic diagram of the direction of the external in-plane electric field. **b** Schematic diagram of a four-terminal Hall-bar-like device. **c** Longitudinal and **d** transverse spin conductivities of

RuF₄ as a function of ϕ . **e** Longitudinal and **f** transverse spin conductivities of CuF₂ as a function of ϕ . Fermi level is set to -0.1 eV in (c–f).

ferroelastic altermagnet, enabling information storage. By applying a small read voltage and measuring the channel resistance – where each strain state maps to a distinct resistance value – the stored multi-level information can be read out through discrimination of these resistance levels.

Discussion

In summary, we proposed the concept of ferroelastic altermagnets that synergizes ferroelasticity with altermagnetism. The switchable ferroelastic states in ferroelastic altermagnets drive multistate nonvolatile switching of the altermagnetic spin splitting, leading to an *altermagnetoelastic effect*. Using monolayers RuF₄ and CuF₂ as proof-of-concept, 2-state and 3-state nonvolatile switching of the altermagnetic spin splitting are demonstrated. We further show that the multistate ferroelastic phase transitions are accompanied by a switching of the spin current associated with the altermagnetic spin splitting. We note that the ferroelastic altermagnetism can also be achieved in other 2D ferroelastic altermagnets such as the two-state ferroelastic monolayers of VF₄^{11,13,66,67} and OsF₄¹³, and the three-state ferroelastic monolayer AgF₂^{13,60,68}.

Methods

First-principles calculations

First-principles calculations are performed based on density functional theory (DFT)⁶⁹ as implemented in Vienna ab initio simulation package (VASP)⁷⁰. Exchange-correlation interaction is described by the Perdew-Burke-Ernzerhof (PBE) parametrization of generalized gradient approximation (GGA)⁷¹. Structures are relaxed until the force on each atom is less than 0.01 eV/Å. The cutoff energy and electronic iteration convergence criterion are set to 600 eV and 10^{-6} eV, respectively. To sample the 2D Brillouin zone, Monkhorst-Pack (MP) k-grid meshes⁷² of $7 \times 7 \times 1$ and

$5 \times 9 \times 1$ are used for monolayers RuF₄ and CuF₂, respectively. To avoid the interaction between adjacent layers, a vacuum space of 20 Å is added. Phonon calculations are carried out using the PHONOPY code with a $4 \times 5 \times 1$ and $4 \times 4 \times 1$ supercell for monolayers RuF₄ and CuF₂, respectively⁷³. Ab initio molecular dynamics (AIMD) simulations are performed at 300 K for 5 ps with a time step of 1 fs with a $4 \times 5 \times 1$ and $4 \times 4 \times 1$ supercell for monolayers RuF₄ and CuF₂, respectively. Ferroelastic switching barrier is obtained by nudged elastic band (NEB) method⁷⁴. The spin-resolved transport properties are calculated using a housing-made code, in which the electron energy and electron group velocity are evaluated from the Wannier-based tight binding Hamiltonian⁷⁵.

Data availability

All study data is included in the article.

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Author contributions

Y.S.A. and R.P. conceived and designed the project. R.P. performed the calculations and analysis. T.Z. reviewed and analyzed the results. Y.S.A. and R.P. wrote the manuscript with inputs from all authors.

Competing interests

The authors declare no competing interests.

Additional information

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